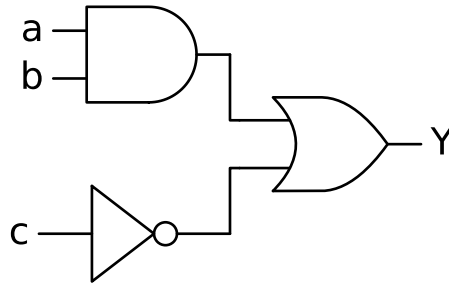


Boolean Algebra Worksheet

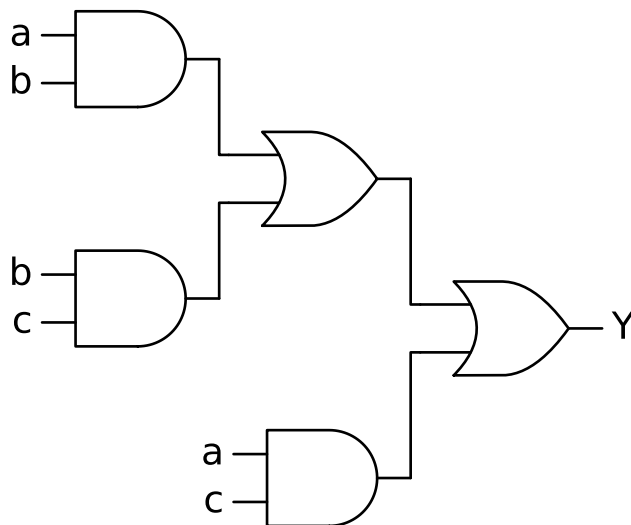
Reminder: $A \oplus B = \overline{A}B + A\overline{B}$ (XOR)

Problem 1.



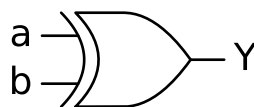
Write the Boolean expression for output Y , then draw its truth table.

Problem 2.



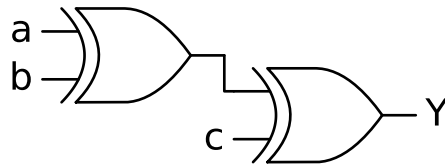
Write the Boolean expression for output Y , then draw its truth table.

Problem 3.



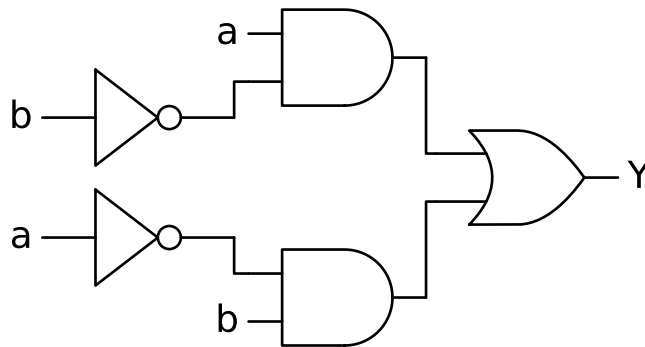
Write the Boolean expression for output Y , then draw its truth table.

Problem 4.



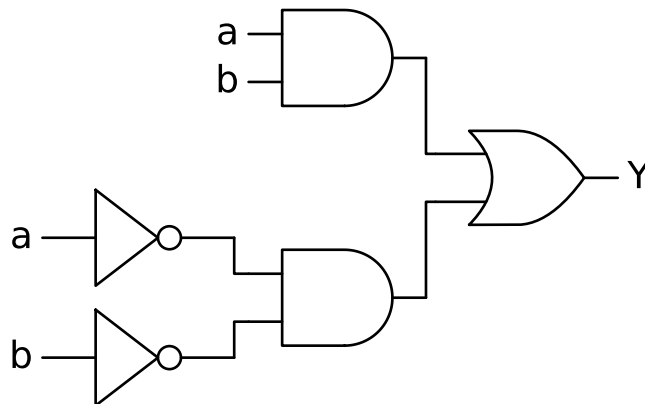
Write the Boolean expression for output Y , then draw its truth table.

Problem 5.



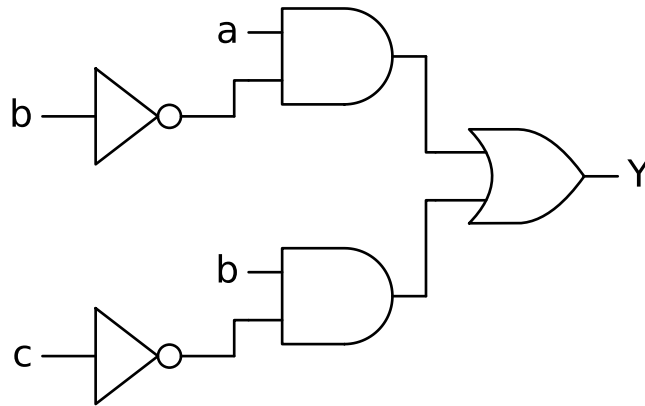
Write the Boolean expression for output Y , then draw its truth table.

Problem 6.



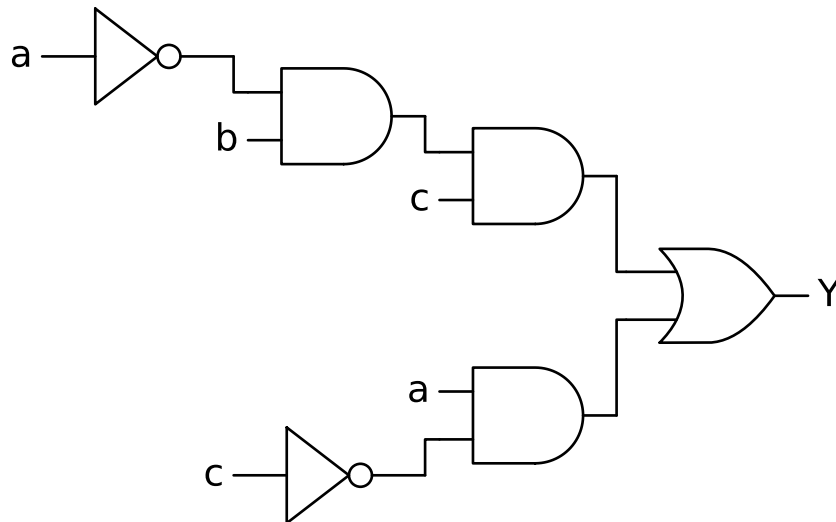
Write the Boolean expression for output Y , then draw its truth table.

Problem 7.



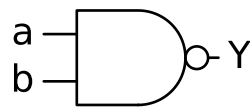
Write the Boolean expression for output Y , then draw its truth table.

Problem 8.



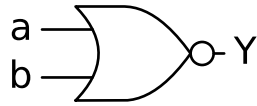
Write the Boolean expression for output Y , then draw its truth table.

Problem 9.



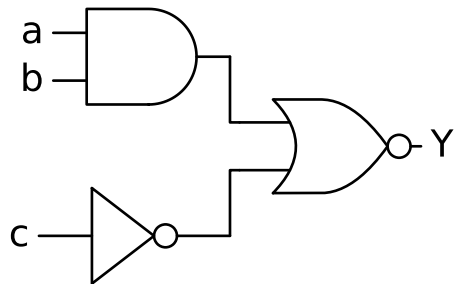
Write the Boolean expression for output Y , then draw its truth table.

Problem 10.



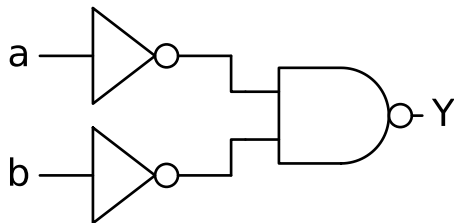
Write the Boolean expression for output Y , then draw its truth table.

Problem 11.



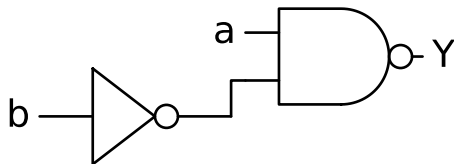
Write the Boolean expression for output Y , then draw its truth table.

Problem 12.



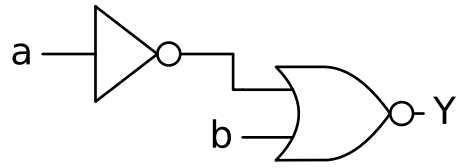
Write the Boolean expression for output Y , then draw its truth table.

Problem 13.



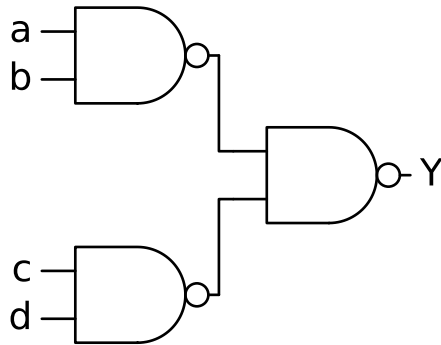
Write the Boolean expression for output Y , then draw its truth table.

Problem 14.



Write the Boolean expression for output Y , then draw its truth table.

Problem 15.



Write the Boolean expression for output Y , then draw its truth table.

Problem 16.

Draw the logic-gate circuit for:

$$Y = A \cdot (B + C)$$

Problem 17.

Draw the logic-gate circuit for:

$$Y = \overline{A \cdot B} + C$$

Problem 18.

Draw the logic-gate circuit for:

$$Y = (A + B) \cdot (A + C)$$

Problem 19.

Draw the logic-gate circuit for:

$$Y = A \oplus B \cdot C$$

Problem 20.

Draw the truth table for:

$$Y = A \cdot B + C$$

Problem 21.

Draw the truth table for:

$$Y = A \oplus B \oplus C$$

Problem 22.

Draw the truth table for:

$$Y = \overline{A + B} \cdot C$$

Problem 23.

Draw the truth table for:

$$Y = A \cdot \overline{B} + \overline{A} \cdot C$$
